



## MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BS-M101/BSM101 Mathematics -IA

UPID : 001004

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

## Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[ 1 × 10 = 10 ]

- (i)  $f(x) = |x|$  satisfies Roll's theorem  $[-1, 1]$ . State true or false with reasons.
- (ii) Every square matrix  $A$  can be expressed as a sum of symmetric and skew symmetric matrix and it can be expressed as=
- (iii) Collection of vectors containing null vectors is linearly \_\_\_\_\_.
- (iv) The eigen value of the matrix  $A$  are  $a$  and  $b$ ; then what are the eigen values of  $A^2$ ?
- (v) The value of  $\Gamma(5/2) = ?$
- (vi)  $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x} = ?$
- (vii) Find the value of  $C$  for which the vector  $(1, 2)$  and  $(3, C)$  linearly independent.
- (viii) A set of vectors in a vector space  $V$  over a field  $F$  is orthogonal if  $\| \alpha \| = \underline{\hspace{2cm}}$ .
- (ix) What is eigen value of a square matrix?

Find the eigen value of the matrix  $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

- (x) Define Beta and Gamma functions.
- (xi) The product of two Orthogonal matrix is \_\_\_\_\_ matrix.
- (xii) The product of any matrix and its transpose is \_\_\_\_\_ matrix.

## Group-B (Short Answer Type Question)

Answer any three of the following :

[ 5 × 3 = 15 ]

2. Check whether the vectors  $\{ (1, 1, 0), (1, 1, 1), (0, 1, -1) \}$  is a basis in  $R^3$  or not. [5]
3. State Lagrange's mean value theorem. [5]

Prove that  $\frac{x}{1+x} < \log(1+x) < x, x > 0$

4. Solve the following equations by matrix inversion method, if possible: [5]

$$2x+3y+z=6; 3y+5z=8; -x+y=0$$

5. Define subspace of a vector space. [5]

Let  $V = \{ M_2 \text{ set of all } 2 \times 2 \text{ matrices of the form } \begin{pmatrix} a & b \\ c & d \end{pmatrix}, a, b, c, d \in R \}$  and  $S$  is the subset of  $V$  such that  $S = \{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a + b = 0 \}$  is a subspace of  $V$ . Prove that  $S$  is a subspace of  $V$ .

6. Prove that  $B(m, n) = B(n, m)$ . Also write the relation between Beta and Gamma functions and find  $B(3, 4)$ . [5]

## Group-C (Long Answer Type Question)

Answer any three of the following :

[ 15 × 3 = 45 ]

7. (a) Define kernel of a linear transformation between two vector spaces. [5]

Is  $f(x) = \sin x$  a Linear transformation? Justify your answer.

- (b) State Cayley Hamilton theorem for matrices. What is the Characteristic equation of a square matrix? Explain with an example. [4]

- (c) What is eigen vector? [6]

Find the eigen values and eigen vectors of matrix  $\begin{pmatrix} 2 & -1 \\ 1 & 4 \end{pmatrix}$ .

8. Show that the set of all complex numbers is a vector space over real field w.r.t the usual addition between two complex number and multiplication between a real number and a complex number. Find a basis of this space and hence the dimension. [ 15 ]
9. What is diagonalization of matrix? [ 15 ]

Diagonalise the matrix  $A = \begin{pmatrix} 6 & 4 & -2 \\ 4 & 12 & -4 \\ -2 & -4 & 13 \end{pmatrix}$ . Write the matrix P, such that  $P^{-1}AP$  is diagonal matrix. Also show that  $P^{-1}AP = D$ .

10. (a) Prove that  $\int_0^\infty \frac{1}{(1+x^2)^5} dx = 35\pi/256$  [ 7 ]
- (b) Obtain the evolute of the parabola  $y^2 = 4ax$ . [ 8 ]
11. (a) Verify Lagrange's MVT for  $f(x) = x(x-1)(x-2)$  in  $[0, 1/2]$ . [ 5 ]
- (b) Expand the following functions in power of x in infinite series stating in each case the condition under which the expression is valid:  $f(x) = \cos x$ . [ 5 ]
- (c) Evaluate  $\Gamma(1/2) \Gamma(3/2) \Gamma(5/2) \Gamma(7/2)$ . [ 5 ]

\*\*\* END OF PAPER \*\*\*